Testing & Validation Framework
STM32-Based Synchronous Buck Power Supply

Variable Switching DC Power Supply | 1.25–30 V | 0–5 A | >88% Efficiency

Phase 0 — Design Validation (Before PCB Fabrication)

Goal: Catch errors while they are still free to fix.

Test	Method	Acceptance Criteria
Schematic Review	Peer review with checklist: pinouts, decoupling, protection diodes, bootstrap cap polarity, current shunt placement	No open items
SPICE Simulation	Run buck_powerstage.cir across corners: min Vin (30 V), nominal (45 V), max (50 V); min/max load; temp extremes	Output stable, ripple < 30 mVpp, no sub-harmonic oscillation
Worst-Case Analysis (WCA)	Calculate max inductor current (including ripple), MOSFET Vds stress, capacitor ripple current, bootstrap cap droop	All components derated to ≤70% of absolute max
Firmware Review	Static analysis of PID clamping, division-by-zero checks, ISR execution time budget	ISR < 45 µs (leaves 5 µs headroom in 50 µs slot)

Why this matters: A schematic error caught here costs zero dollars. Caught after assembly, it costs a board spin, components, and days of debug.

Phase 1 — Incoming Inspection (Board Arrival)

Goal: Verify the factory built what you designed.

Test	Method	Tool
Visual Inspection	Check for solder mask voids, trace shorts, drill alignment, component footprints	Magnifier / AOI
Netlist Verification	Beep-test critical nets: 3.3 V rail, GND, SW node to inductor, gate drive to MOSFETs	DMM in continuity mode
Isolation Test	Verify >10 MΩ between AC primary (before T1) and DC output at 500 VDC	Hipot / Insulation tester
Impedance Check	VDD–GND resistance > 100 Ω; no shorts on gate drive lines	DMM

Phase 2 — Low-Voltage Digital Bring-Up (No Mains)

Goal: Prove the brain works before connecting the brawn. Setup: Power the STM32 + control section from a current-limited 5 V bench supply (200 mA limit).

Test	Procedure	Pass Criteria
Power Sequencing	Apply 5 V, measure 3.3 V LDO output	3.30 V ± 5%
Current Draw	Measure total 5 V input current	< 100 mA idle
Clock & Reset	Probe 8 MHz crystal (if used) or verify HSE lock	Stable clock, no brownouts
I2C Bus Scan	Run i2cdetect-equivalent or logic analyzer on PB6/PB7	Detects INA226 (0x40) and OLED (0x3C)
OLED Initialization	Power cycle, observe	Correct display, no

	startup screen	garbled pixels
Encoder Input	Rotate encoder, monitor PA6/PA7 on scope	Quadrature pulses present
GPIO Sanity	Toggle PA4 (Fault LED), PC13 (CV), PC14 (CC)	LEDs respond to firmware commands
ADC Sanity	Apply known voltage to PA0 via divider, read raw ADC counts	Linear response, no stuck bits

Gate: Do not proceed to Phase 3 until all digital I/O is verified. Debugging SPI/I2C with 45 V on the bus is dangerous and distracting.

Phase 3 — Open-Loop Power Stage Bring-Up (DC Injection)

Goal: Validate the powertrain without the risk of mains energy.

Setup: Disconnect T1 secondary. Feed the bulk capacitor node from a current-limited bench supply (e.g., 15 V, 200 mA limit). Do not connect the STM32 feedback loop yet — drive PA8 manually or with a function generator.

Stage	Voltage	Current Limit	What to Verify
1	15 V	200 mA	No smoke; 3.3 V rail stays up; SW node shows switching waveform
2	30 V	500 mA	Output follows duty cycle; inductor current triangular; no shoot-through
3	45 V	1 A	Full voltage stress; MOSFETs switch cleanly; bootstrap cap

			holds charge

Specific checks at each stage:

Test	Method	Pass Criteria
Gate Drive Waveform	Scope HO and LO (use isolated probe or differential method)	Clean 10–15 V amplitude; rise/fall < 100 ns; ~520 ns dead-time visible
SW Node	Probe between Q1 source / Q2 drain	Square wave 0 → Vin; no ringing > 20% of Vin; no DC shoot-through
Bootstrap Diode	Scope VB (pin 8 of IR2104) vs. VS (pin 6)	VB rides ~12 V above VS; refreshes every cycle
Inductor Current	Current probe in series with L1	Triangular ripple; peak < 6.8 A (Isat limit); average matches load
Diode Emulation	Remove gate drive to Q2, let body diode freewheel	Output drops but remains stable — confirms non-synchronous fallback works

Why staged voltage: At 15 V, a shoot-through fault dissipates P = V×I = 15 V × 200 mA = 3 W — survivable. At 45 V with a 4700 µF capacitor behind it, the same fault dumps hundreds of joules instantly.

Phase 4 — Closed-Loop Control Validation

Goal: Prove the PID loop regulates and protects.

Setup: Reconnect STM32 feedback. Keep bench supply on DC bus. Load with electronic load or resistor bank.

Test	Procedure	Pass Criteria
Soft Start	Command 12 V setpoint from 0 V	Output ramps monotonically; no overshoot > 10%

Setpoint Tracking	Step setpoint: 1.25 V → 5 V → 12 V → 24 V → 30 V	Each level settles within ±0.5%; no limit cycling
PID Parameter Sweep	Vary Kp, Ki, Kd via UART/debug interface	Identify stable region; confirm chosen values are not on the edge of instability
ISR Timing	Toggle spare GPIO (e.g., PB12) in PID ISR; scope it	20 kHz ± 1%; jitter < 1 µs
ADC Sampling	Scope ADC start trigger and PWM update	Sample occurs at quiet point (e.g., mid-switching cycle if sync'd)

Phase 5 — Performance Characterization

Goal: Map the supply's behavior across its entire operating envelope.

5A — Static Performance

Parameter	Test Points	Tool
Load Regulation	12 V @ 0 A, 1 A, 2.5 A, 5 A	DMM + E-load
Line Regulation	Vary input 35 V → 45 V → 50 V at 12 V / 2.5 A	DMM
Cross-Regulation	Test all output voltages (1.25 V, 5 V, 12 V, 24 V, 30 V) at min, mid, max load	DMM
Ripple & Noise	All voltage/load corners; AC-coupled, 20 MHz BW, ground spring technique	≥100 MHz scope

5B — Dynamic Performance

Test	Method	Pass Criteria
Load Transient	0 ↔ 5 A step at 12 V; 10–90% step time <	Settling < 1 ms; overshoot < 5%

	10 µs	
Line Transient	Step input 40 V ↔ 50 V at full load	Output deviation < 2%; recovery < 500 µs
Startup into Load	Enable output with 5 A load already applied	No foldback latch-up; clean start

5C — Efficiency Mapping

Efficiency vs. Load: At 5 V, 12 V, 24 V output, sweep 0.5 A → 5 A in 0.5 A steps. Measure Pin (input voltage × input current) and Pout (output voltage × output current). Plot η = Pout/Pin.

Loss Breakdown: Estimate: Pcond (I^2Rds), Psw (switching), Pinductor (DCR), Pcap (ESR). Compare sum to measured loss.

Target: >88% at 12 V / 5 A. If you measure 85%, the loss breakdown tells you whether to swap MOSFETs (lower Rds), upgrade the inductor (lower DCR), or reduce switching frequency.

Phase 6 — Protection & Fault Injection

Goal: Prove the supply fails safely.

Fault	Injection Method	Expected Response	Measurement
Over-Voltage (OVP)	Override voltage sense divider with external voltage; or command setpoint above limit	/SD asserts (PB12 low); PWM stops; output collapses	Scope /SD and Vout; verify < 20 µs response
Over-Current (OCP) — Foldback	Increase E-load current beyond 5 A	Output transitions from CV to CC mode; voltage folds back while	Verify 5.0–5.3 A foldback point

		current holds at limit	
Over-Current (Fast) — ALERT	Step E-load to >6 A suddenly	INA226 ALERT → EXTI0 (PB0) → ISR disables PWM via /SD	Scope PB0 and Vout; verify < 10 µs
Short Circuit	Hard short output with relay or MOSFET	Immediate shutdown; no damage upon removal of short	Thermal check MOSFETs after 10 cycles
Over-Temperature	Heat NTC with hot air gun to >80 °C	Firmware throttles or shuts down	Verify graceful degradation, not abrupt latch
Reverse Polarity (Output)	Connect reverse voltage to output terminals	Body diode of Q2 conducts; verify fuse or ideal diode protection if present	Check for damage
Input Undervoltage	Drop bench supply to < 30 V	UVLO triggers; output shuts down cleanly	No erratic behavior near dropout

Astranis-relevant note: For flight hardware, fault testing is exhaustive. You'd run each fault 50+ times across temperature corners to prove deterministic behavior. For your bench supply, 5–10 cycles per fault is a good baseline.

Phase 7 — Environmental & Stress Testing

Goal: Prove robustness beyond nominal conditions.

Test	Method	Duration / Cycles	What to Watch
Thermal Cycling	Run in environmental chamber: 0 °C ↔ 50 °C, powered and loaded	10 cycles	Output drift, cold-start reliability, solder joint integrity

High-Temp Soak	50 °C ambient, full load	4 hours	Thermal runaway? Fan control (PB9) responds?
Vibration (Lite)	Random vibration 5–20 Hz, 0.5 g (if available)	10 min per axis	Connector loosening, inductor/magnetics mechanical resonance
Burn-In	30 °C, 80% load, continuous operation	48 hours	Early-life failures, capacitor ESR drift
EMI Pre-Compliance	Near-field probe around switching node; line impedance stabilization network (LISN) on AC input if available	Spot checks	Compare to your 118 mVpp baseline; identify radiating loops

Phase 8 — Documentation & Release

Goal: Make the test repeatable and the design transferable.

Deliverable	Contents
Test Report	Test date, equipment serial numbers/cal dates, ambient conditions, raw data tables, pass/fail per requirement
Calibration Certificate	ADC voltage sense: apply 5.000 V, record ADC counts, derive correction factor; repeat for current sense
Known Issues / Limitations	e.g., "OLED refresh causes 2 µs jitter in PID ISR; acceptable for bench use"
Regression Test Script	Python script (PyVISA) that automatically runs Tests 4–7 and

	logs to CSV

Summary: The "Gate" Structure

A phase-gated approach ensures no stage is entered until the previous one is fully validated.

Gate	Condition to Proceed
G0	Simulation and WCA complete; no open items
G1	Incoming inspection pass; no shorts; isolation verified
G2	Digital bring-up complete; all peripherals respond
G3	Open-loop power stage verified at 15 V, 30 V, 45 V; no shoot-through
G4	Closed-loop regulation stable across all setpoints
G5	Performance meets spec: regulation, ripple, efficiency
G6	All protection faults injected and passed
G7	Environmental stress complete; no degradation
G8	Documentation complete; design released